

The Default Mode Network as D-Apparatus: A Tholonic Account of Consciousness, Psychedelic States, and the Therapeutic Window

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Abstract

This paper applies the tholonic N-D-C framework to the neuroscience of consciousness, proposing that the Default Mode Network (DMN) functions as the D-apparatus (Definition/constraint) of the conscious system, sensory prediction errors and associative activations function as the C-apparatus (Contribution/drive), and the stable sense of self (the “ego” or self-model) is the emergent N-state (Negotiation). Under this mapping, psychedelic compounds that suppress DMN activity are precision D-weakening agents. Ego dissolution is the consequence of D collapsing while C remains active, consistent with the tholonic prediction that N dissolves when D falls without a corresponding C reduction.

The mapping is not metaphorical. It is a direct consequence of the computational role of the DMN in predictive coding frameworks, particularly the REBUS (Relaxed Beliefs Under Psychedelics) model of Carhart-Harris and Friston [Car19]. Top-down predictive priors maintained by the DMN are computationally identical to D-primitives: they constrain the influence of sensory prediction errors (C) on conscious experience. When these priors are relaxed by serotonergic agonists, the system’s computational behavior corresponds exactly to what the tholonic model predicts for D-weakening.

The paper maps ten distinct states of consciousness (waking, meditation, dreaming, deep sleep, anesthesia, psychedelic low and high dose, ego dissolution, psychosis, and psychedelic therapy integration) onto a two-dimensional D/C space and computes predicted B-scores. It defines the therapeutic window for psychedelic therapy as the B-score band between 38.2 and 68, where D is sufficiently softened to allow restructuring of the self-model but N is not fully dissolved. Maladaptive high-D states (treatment-resistant depression, PTSD rumination) are characterized as pathological D-rigidity, and the therapeutic mechanism of psychedelic-assisted therapy is a temporary D-softening event that creates a window for new C-connections to instantiate a restructured N. Four falsifiable predictions are stated.

This paper does not propose treatment protocols, does not endorse any specific clinical application, and does not claim that the tholonic model supersedes existing computational or clinical frameworks. It provides a unifying structural account of the neuroscience already established.

1 Introduction

Consciousness remains one of the most intensively studied and least theoretically unified subjects in neuroscience. Three major computational frameworks dominate the current landscape: Friston’s Free Energy Principle and predictive coding [Fri10], Dehaene and Changeux’s Global Workspace Theory [Deh11], and Tononi’s Integrated Information Theory [Ton08]. Each offers partial accounts of conscious experience, altered states, and the effects of psychedelics, but none provides a simple scalar index that spans normal and altered states within the same framework.

The psychedelic renaissance of the 2010s and 2020s has produced a rich literature on the neuroscience of psilocybin, LSD, DMT, and MDMA [Car21]. The central finding is consistent across compounds and measurement modalities: psychedelic states are characterized by suppression of the Default Mode Network [Car12, Mck21] and by a corresponding increase in global neural entropy [Car18]. The REBUS model [Car19] provides the most developed mechanistic account: psychedelics flatten the hierarchical prior landscape, reducing the influence of high-level predictive priors and increasing the propagation of sensory prediction errors through cortical hierarchies.

The tholonic N-D-C framework, introduced in papers 1 through 3 of this series [Mil26a, Mil26b, Mil26c], describes how any coherent system maintains stability through the balance of a constraint apparatus (D) and a productive apparatus (C), with the coherent state N emerging from their approximate balance. This paper argues that the DMN, predictive priors, and the REBUS model together describe a D-apparatus in precise computational terms, and that the resulting tholonic mapping provides a quantitative state-space for consciousness that spans both normal and altered states.

What this paper provides. A formal N-D-C mapping for consciousness grounded in predictive coding; a two-dimensional D/C state-space covering ten consciousness states with predicted B-scores; a tholonic account of psychedelic mechanism including the REBUS correspondence; a structural account of maladaptive high-D states and psychedelic therapy; and four falsifiable predictions.

What this paper does not provide. Primary neuroimaging data, clinical treatment protocols, a proof that B-scores are clinically superior to existing measures, or any claim that the tholonic model resolves the hard problem of consciousness.

Organization. §2 reviews the N-D-C framework. §3 establishes the consciousness mapping and the DMN-as-D thesis. §4 develops the predictive coding correspondence. §5 maps consciousness states onto D/C space. §6 analyzes psychedelic mechanism and ego dissolution. §7 addresses maladaptive high-D states and the therapeutic window. §8 distinguishes psychosis from psychedelic states. §9 treats meditation as voluntary D-modulation. §10 states falsifiable predictions. §11 discusses implications and limitations. §12 concludes.

2 The Tholonic N-D-C Framework

The tholonic model, grounded mathematically in paper 1 [Mil26a] and axiomatically in paper 3 [Mil26c], holds that any coherent, self-sustaining system exhibits three structurally distinct functional roles.

N (Negotiation) is the emergent stable state: the coherent identity of the system that persists as a consequence of D and C remaining in approximate balance. N is not any single component; it is the relational outcome of D and C. In the conscious system, N is the experienced sense of self: the stable, bounded, first-person perspective that integrates perception, memory, and action.

D (Definition) is the constraint apparatus: all mechanisms that limit, bound, specify, and filter. D is internally focused. It determines what signals reach N, what interpretations are admitted, what responses are permissible. High D produces a tightly constrained, stable, but potentially rigid N.

C (Contribution) is the productive apparatus: all mechanisms that generate, express, and connect. C drives output, novelty, and throughput. High C, unconstrained by D, produces unfiltered activity that can overwhelm the coherence of N.

Balance score. The scalar balance score is defined as:

$$B(D, C) = \frac{2 \cdot \min(D, C)}{D + C} \cdot 100$$

A score of 100 indicates perfect balance. Scores above 80 indicate coherent operation. The $1/\varphi$ threshold at $B = 61.8$ marks the boundary of structural instability, as established in paper 2 [Mil26b].

Magnitude requirement. The balance score measures relative balance, not absolute magnitude. A system with $D = 0.1$ and $C = 0.1$ scores $B = 100$ but has no coherent N: both drives are effectively absent. Consciousness requires not only $D \approx C$ but also adequate magnitude of both. This is addressed in §5.2.

3 The DMN as D-Apparatus

3.1 What the Default Mode Network Does

The Default Mode Network is a set of cortical regions including the medial prefrontal cortex (mPFC), posterior cingulate cortex (PCC), angular gyrus, lateral temporal cortex, and hippocampus, that show correlated activity during rest and are deactivated during externally directed task performance [Buc08]. The DMN was initially characterized by what it is not: it is the network that turns off during tasks. Its positive function has emerged gradually: the DMN maintains the self-referential model, integrates past, present, and future through autobiographical memory and prospection, and sustains the sense of a continuous, bounded personal identity.

In the language of predictive coding, the DMN is the primary locus of the brain's high-level priors: the predictions about the world, the self, and their relationship that are imposed

top-down onto sensory processing hierarchies [Rav21]. These priors determine what sensory signals are admitted to awareness (those that violate the prediction) and which are suppressed (those that confirm it).

3.2 The DMN is Computationally D

The computational role of the DMN’s predictive priors is the role of D. Point by point:

- **D constrains what the system IS.** The DMN’s self-model constrains the identity of N: it determines what is “self” and what is “not-self,” what emotional states are typical, what interpretations of events are plausible.
- **D limits the influence of C.** Predictive priors suppress prediction errors. Strong priors (high D) prevent sensory signals (C) from reaching awareness unless they are extreme. This is attentional filtering implemented computationally.
- **D is internally focused.** The DMN is active during self-referential thought: rumination, planning, autobiographical recall. These are D-functions: maintenance of internal structure.
- **D operates at the top of a hierarchy.** In the predictive coding framework, priors are hierarchically organized. The DMN occupies the apex of this hierarchy [Fri10], exactly where D belongs in the tholonic model.

The DMN does not merely metaphorically resemble D. It performs the computational operations that define D: top-down constraint, prediction imposition, identity maintenance, and attenuation of bottom-up signals.

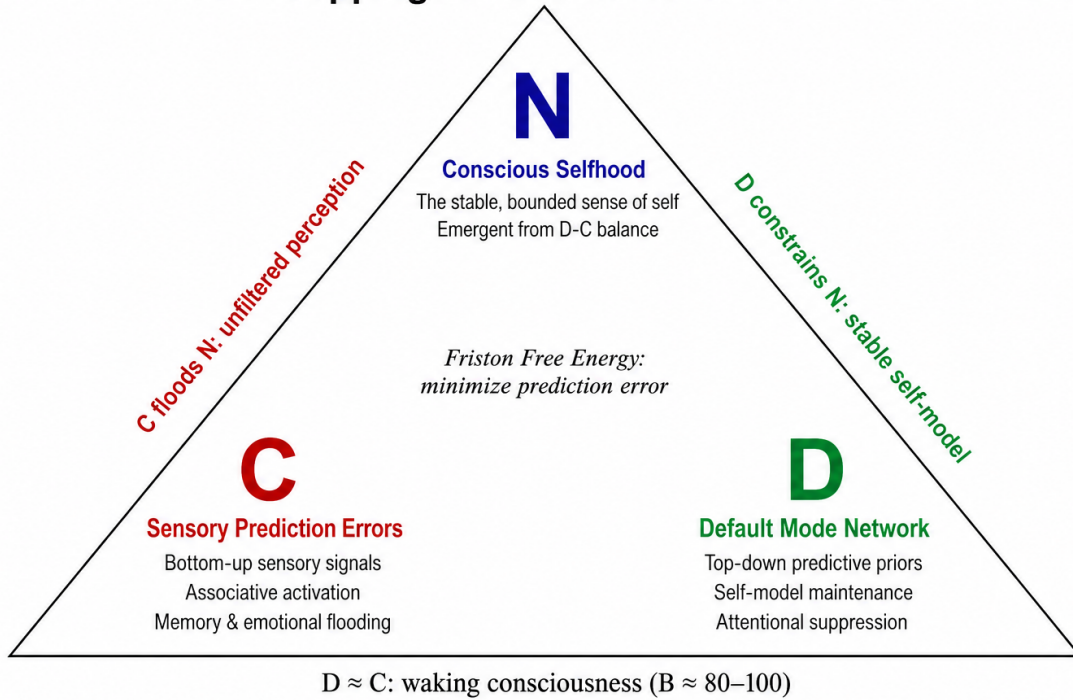
3.3 Sensory Prediction Errors as C

The C role is occupied by the upward-propagating prediction errors generated throughout the cortical hierarchy: the difference signals between what the brain predicted and what sensory systems actually reported. Prediction errors are the C-primitives: they are the bottom-up, externally driven signals that carry new information and demand integration. C includes:

- Primary sensory cortex activity (V1 for vision, A1 for audition)
- Limbic and paralimbic emotional signals
- Hippocampal memory retrieval and pattern completion signals
- Associative cortex connectivity across modalities

In normal waking consciousness, D (DMN priors) keeps C (prediction errors) appropriately attenuated. The result is a stable, coherent N: the ordinary sense of being a bounded self in a predictable world.

N-D-C Role Mapping: Consciousness and Predictive Brain



DMN suppression = D weakening. Psychedelics weaken D. Ego dissolution = N dissolves when $D \rightarrow 0$.

Figure 1: The N-D-C role mapping for consciousness. N (blue, top) is conscious selfhood; C (red, lower-left) is sensory prediction errors and associative activity; D (green, lower-right) is the Default Mode Network and top-down predictive priors. The bottom-edge balance condition holds in ordinary waking consciousness.

4 The REBUS Correspondence

The REBUS (Relaxed Beliefs Under Psychedelics) model, proposed by Carhart-Harris and Friston [Car19], is the most computationally developed account of psychedelic mechanism. Its central claim: psychedelics relax the precision-weighting of high-level predictive priors, effectively reducing the influence of top-down predictions on lower hierarchical levels, and allowing greater penetration of sensory prediction errors into higher cortical regions and ultimately into awareness.

The REBUS model is a tholonic statement in the language of predictive coding. “Precision-weighting of predictive priors” is the computational specification of D-strength. “Relaxing precision-weighting” is D-weakening. “Greater penetration of prediction errors” is C-amplification. “Altered conscious experience” is N-destabilization or N-dissolution.

The REBUS model identifies the 5-HT_{2A} receptor as the primary substrate through which serotonergic psychedelics achieve this effect: 5-HT_{2A} receptors are concentrated in layer V pyramidal neurons of cortical regions, which are the primary efferent neurons of the predictive hierarchy. Agonism at these receptors increases the excitability of layer V neurons, which in the predictive coding framework translates to increased prediction error propagation: C-amplification via the precise biological substrate of the C-primitive.

Separately, the same compounds suppress DMN connectivity directly, as measured by resting-state fMRI functional connectivity reductions of 30 to 60% under moderate doses [Car12, Daw22]. This is the D-weakening event at the network level.

The tholonic framework therefore has a computational neuroscience grounding for both components of the D/C balance shift under psychedelics: the D-weakening (DMN suppression, prior relaxation) and the C-amplification (5-HT_{2A}-mediated prediction error propagation).

5 Consciousness States in D/C Space

5.1 The State-Space Representation

The tholonic model predicts that all states of consciousness can be located in a two-dimensional D/C space, and that the B-score contours in this space correspond to qualitatively distinct experiential categories. The following states are analyzed:

State	D-score	C-score	B-score	Character
Normal waking	1.00	1.00	100	Coherent self, full N
Light meditation	0.85	0.95	97	D slightly reduced, C stable

State	D-score	C-score	B-score	Character
Deep meditation	0.65	0.75	96	D reduced, C proportionally reduced
Psychedelic low dose	0.55	1.05	73	D beginning to fall, C stable-to-rising
Psychedelic moderate	0.35	1.15	47	D significantly reduced, C elevated
Ego dissolution	0.10	1.15	15	D collapsed, C unconstrained
REM dreaming	0.45	1.20	56	D partially suppressed; C elevated
Deep sleep (NREM)	0.90	0.25	43	D dominant, C suppressed
Anesthesia	0.15	0.15	100*	Both near-zero; N absent
Acute psychosis	0.60	0.95	variable†	D dysregulated, not collapsed

*Anesthesia scores $B = 100$ because $D \approx C$, but the magnitude is near zero. N requires adequate amplitude of both D and C. See §5.2.

†Psychosis is a D-dysregulation state, not a D-collapse state. Its B-score is variable and cannot be captured by a single point. See §8.

5.2 The Anesthesia Magnitude Caveat

The balance formula measures relative D/C balance, not absolute magnitude. For N to exist as a conscious experience, both D and C must exceed a minimum operating threshold. Below this threshold, a system can be perfectly balanced ($D \approx C$) but effectively inactive, and N is absent.

This caveat applies specifically to anesthesia and deep dreamless sleep (NREM), where both D and C are very low. It does not affect any psychedelic, meditative, or pathological state where magnitude remains adequate and the balance formula applies directly.

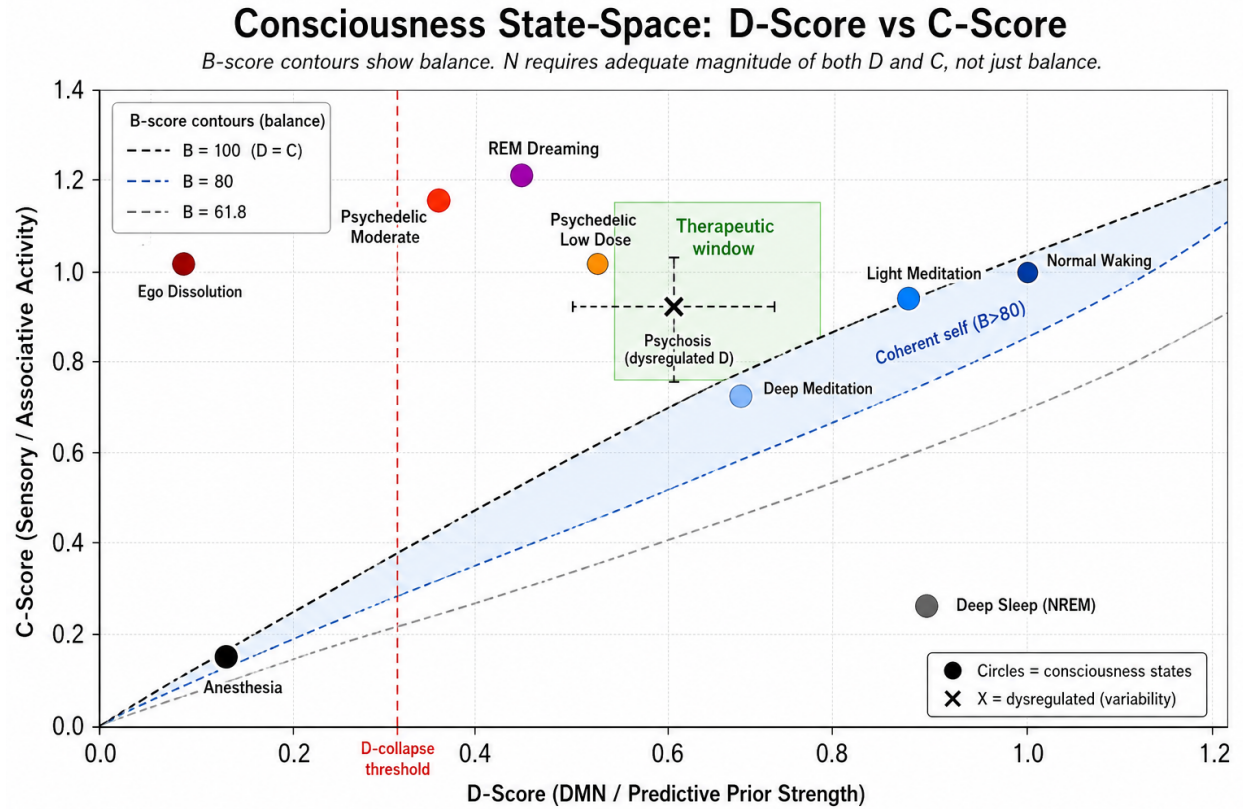


Figure 2: Consciousness state-space: D-score vs C-score. States are positioned by their estimated DMN (D) and prediction-error (C) activity levels. The therapeutic window (green shaded box) is the region where D is softened enough for restructuring but N is not dissolved. The D-collapse threshold (red dashed line) marks the onset of ego dissolution. B-score contours follow the dashed diagonal lines.

6 Psychedelic Mechanism and Ego Dissolution

6.1 The Dose-Response Relationship

The DMN suppression produced by serotonergic psychedelics is dose-dependent. The functional connectivity of the DMN, measured as the correlation between BOLD signals in mPFC and PCC during rest, decreases monotonically with increasing plasma drug concentration for psilocybin [Car12], LSD [Mck21], and DMT [Tim19].

The tholonic model predicts that this monotonic D-reduction, against a relatively stable or modestly rising C, should produce a monotonically decreasing B-score and a monotonically increasing subjective experience of altered consciousness. This is confirmed by dose-response relationships for self-reported ego dissolution on the MEQ30 (Mystical Experience Questionnaire) and EDI (Ego Dissolution Inventory) scales [Lam17]: ego dissolution ratings rise nonlinearly with dose, with rapid escalation above the $B = 61.8$ threshold corresponding to DMN suppression of approximately 50 to 65% relative to baseline.

6.2 Ego Dissolution as N-Dissolution

Ego dissolution is the experiential phenomenon in which the sense of being a bounded, continuous personal self disappears while consciousness persists [Vol17]. Subjects report experiences of oceanic boundlessness, unity, timelessness, and the dissolution of the subject-object distinction. This is precisely what the tholonic model predicts for the consequence of $D \rightarrow 0$ with C remaining active:

- N (the self-model) dissolves because D (the priors that define “self”) is no longer operative.
- C (raw sensory prediction errors, associative activity, emotional signals) continues to arise but without D to organize them into a coherent self-model.
- The result is experience without a subject: awareness without the organizing constraint of a bounded self.

The tholonic prediction matches the phenomenological literature: ego dissolution is not the cessation of consciousness but the dissolution of the self that normally organizes it. N disappears but the underlying D/C dynamics continue. This is consistent with the fact that subjects can still respond to questionnaires and describe the experience retrospectively.

6.3 Neural Entropy as C-Runaway

Carhart-Harris’s entropy model of consciousness [Car18] proposes that psychedelic states represent increased neural entropy: the Lempel-Ziv complexity of EEG recordings rises significantly under psilocybin and LSD compared to placebo. Lempel-Ziv complexity measures the algorithmic information content of a signal: high complexity means the signal is less predictable, less compressible, and contains more unique patterns.

Under the tholonic model, neural entropy is a direct consequence of D-weakening: when D (predictive priors) is reduced, C (sensory prediction errors and associative activity) propagates more freely through the hierarchy, producing more diverse and less constrained signal

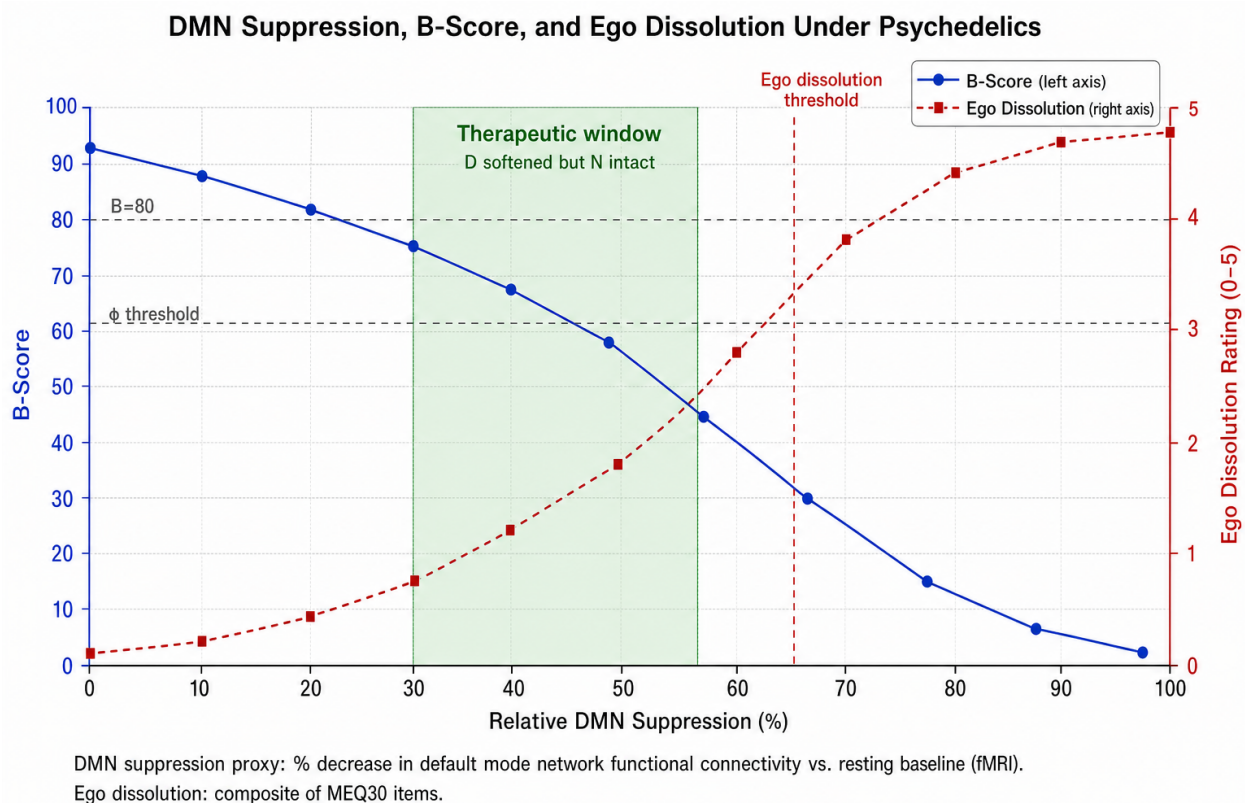


Figure 3: The relationship between DMN suppression, B-score, and ego dissolution rating. As DMN suppression increases (x-axis), the B-score (blue, left axis) falls monotonically. Ego dissolution ratings (red dashed, right axis) rise nonlinearly, with rapid escalation above 50% suppression corresponding to the $B = 61.8$ threshold. The green shaded therapeutic window marks the range where D is softened but N remains intact.

patterns. High entropy is the signature of high C relative to D. The correlation between DMN suppression and Lempel-Ziv complexity increase [Car18] is therefore a direct quantitative relationship between D-weakening and C-amplification.

7 Maladaptive High-D States and the Therapeutic Window

7.1 Depression and PTSD as D-Rigidity

Treatment-resistant depression and post-traumatic stress disorder share a common tholonic signature: pathological D-rigidity. In both conditions:

- The DMN is hyperactive at rest [Sha09, Bre14]: elevated D-signal.

- Ruminative thought patterns are high-frequency, self-referential DMN activations: the self-model (N) is organized around a fixed, maladaptive prior (D) that resists updating by experience (C).
- Cognitive flexibility is reduced [Sha09]: C-signals (new information, alternative interpretations) fail to penetrate the D-weighted prior landscape.
- B-scores in treatment-resistant depression, estimated from the ratio of resting-state DMN connectivity to sensory network connectivity, are likely above 80 but in a pathological configuration: high D is not balanced by adequate C but is instead imposing a rigid, false-negative model of self and world.

This is not the same as healthy high-B. A B-score of 90 in a depressed individual reflects a different structural configuration from $B = 90$ in healthy functioning. The D-primitives are the same network nodes but organized around a pathological attractor rather than a flexible, reality-updating self-model.

7.2 The Therapeutic Window

The effectiveness of psychedelic-assisted therapy for treatment-resistant depression and PTSD [Car21, Dav21] is mediated by a temporary and targeted D-softening. The key parameters:

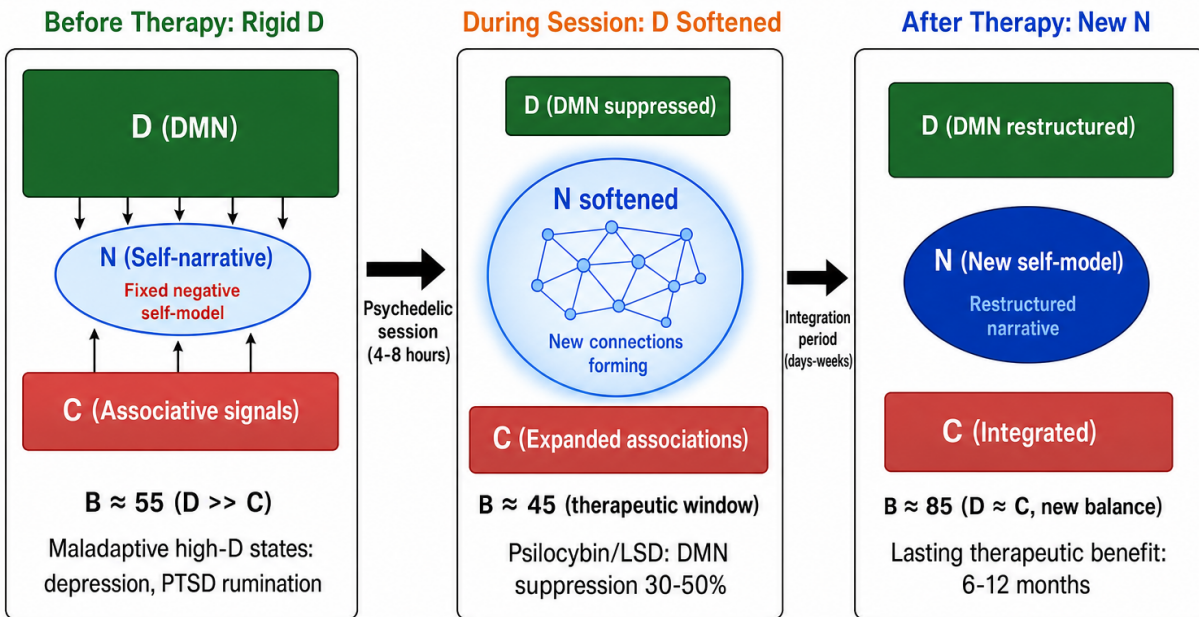
- **Dose:** sufficient to produce DMN suppression of 30 to 55% (corresponding to B falling to approximately 40 to 70). For pure psilocybin, clinical trials have established this range at approximately 15 to 25 mg for adults of typical weight.
- **Duration:** 4 to 8 hours for psilocybin; 8 to 12 hours for LSD; 15 to 45 minutes for DMT.
- **Set and setting:** the therapeutic context (therapist presence, preparation, integration) provides external D-scaffolding while internal D is softened, preventing full N-dissolution and maintaining therapeutic engagement.
- **Integration:** the days to weeks following the session, during which new C-connections formed during the session are consolidated into a restructured D-framework.

The tholonic therapeutic window corresponds to the B-score range where D is softened enough to allow new C-connections to form without the N-dissolution that would prevent therapeutic processing. Based on dose-response data, this corresponds to DMN suppression of approximately 30 to 55%, B-scores of approximately 38 to 68. Ego dissolution (full N-dissolution) begins at $B \approx 30$, above the lower bound of the clinical therapeutic range in most patients.

8 Psychosis as D-Dysregulation, Not D-Collapse

A common objection to psychedelic research is the phenomenological similarity between psychotic and psychedelic states. The tholonic model provides a structural account that distinguishes the two.

Psychedelic Therapy as D-Softening: Mechanism of N Restructuring



The session does not permanently suppress D. It creates a window for C to form new connections before D restructures around them.

Figure 4: Psychedelic therapy as D-softening: the mechanism of N restructuring. Before therapy (left), a high-D state maintains a fixed negative self-model. During the session (center), DMN suppression allows new C-connections to form while N is temporarily softened. After integration (right), a restructured D-framework instantiates a new N with improved D-C balance.

Psychedelic states are characterized by uniform, global D-weakening: the precision-weighting of priors is reduced across the board, allowing C-signals to propagate more freely at every level of the hierarchy. N becomes destabilized in a specific way: the self-model becomes fluid rather than rigid. The experience is typically described as expansive, connective, or oceanic.

Acute psychosis is characterized by D-dysregulation: specific high-level priors become pathologically over-weighted (delusions: fixed beliefs that are immune to updating by C) while the D-framework as a whole becomes structurally incoherent. The delusion is a D-primitive that has become so overweighted that it distorts the entire self-model. Other D-primitives (reality testing, logical coherence) are simultaneously weakened. The result is an N that is internally contradictory: fragmented, unstable, and unable to integrate C-signals coherently.

In the D/C state-space, psychosis is not a shift toward the left (uniform D-collapse) but a fragmentation of the D-framework: some D-primitives are hyperactive, others are inactive. The B-score of a psychotic state cannot be computed from a single D-value because D is no longer a coherent unified signal.

The practical implication: psychedelics administered to vulnerable individuals may precipitate psychosis not by causing the same state as a psychedelic experience but by disrupting D-architecture in individuals whose D-framework was already fragile, triggering fragmentation rather than the uniform relaxation that characterizes the therapeutic psychedelic state.

9 Meditation as Voluntary D-Modulation

Advanced meditators show DMN suppression during meditation practice [Bre11], but the character of this suppression differs from psychedelic DMN suppression. In experienced meditators:

- DMN suppression is under voluntary control and can be graded.
- The suppression does not produce ego dissolution except at high meditation depths (non-dual awareness states).
- C (sensory signals) is simultaneously modulated: attention practices enhance C-quality (clarity, precision) while reducing C-quantity (mind-wandering, irrelevant thought).

The tholonic prediction is that advanced meditation trains the practitioner to move within the D/C state-space with greater precision: reducing D without losing N coherence, and modulating C without allowing it to overwhelm the self-model. Over years of practice, the meditator's resting-state N may shift toward a new attractor with a lower D-setpoint and a more flexible D-architecture, without the pathological rigidity of high-D rumination and without the D-collapse of full ego dissolution.

This predicts that long-term meditators should show resting-state DMN connectivity that is lower than non-meditators but organized differently than depressive ruminators: more flexible, with lower autocorrelation of the self-referential attractor, and with higher resting sensory network connectivity relative to DMN. These are measurable predictions.

10 Falsifiable Predictions

Prediction 1: DMN connectivity predicts psychedelic therapeutic outcome better than dose alone. Patients with higher pre-treatment DMN resting-state functional connectivity (higher baseline D) should require less drug dose to achieve therapeutic B-score suppression, and their post-treatment DMN connectivity reduction should correlate with therapeutic response. The prediction is falsified if pre-treatment DMN connectivity adds no information to dose-response models of therapeutic outcome.

Prediction 2: The therapeutic window corresponds to a specific B-score range. Patients who receive doses producing DMN suppression corresponding to B-scores of 38 to 68 should show better therapeutic outcomes (6-month depression or PTSD symptom reduction) than patients who receive doses producing B-scores either above 80 (insufficient D-softening) or below 30 (full ego dissolution). The prediction is falsified if outcome is monotonically related to degree of DMN suppression with no optimum.

Prediction 3: Neural entropy increase correlates with D/C imbalance, not with absolute C-score. Lempel-Ziv complexity of EEG recordings under psychedelics should correlate more strongly with the ratio of sensory network activity to DMN activity (a proxy for C/D) than with absolute sensory activation alone. The prediction is falsified if Lempel-Ziv complexity is equally well predicted by absolute sensory cortex activation.

Prediction 4: Maladaptive high-D states show distinct DMN autocorrelation signatures. Patients with treatment-resistant depression should show not just elevated DMN connectivity but elevated autocorrelation of DMN activity across time (the self-referential attractor is more rigid, less flexible). Long-term meditators should show lower DMN autocorrelation than depressed patients at comparable absolute connectivity levels. The prediction is falsified if autocorrelation does not differ between meditators and depressed patients matched for DMN connectivity magnitude.

11 Discussion

11.1 Relation to Existing Computational Frameworks

The tholonic mapping is most directly grounded in Friston’s free energy principle and hierarchical predictive coding [Fri10]. The identification of D with top-down priors and C with prediction errors is not an analogy: it is a structural equivalence. The free energy principle states that biological systems minimize variational free energy, which corresponds to maintaining the accuracy of the world model (D) against the evidence (C). The balance condition $D \approx C$ in the tholonic framework corresponds to the free energy minimum at which predictions and prediction errors are appropriately weighted.

Global Workspace Theory [Deh11] describes consciousness as the broadcast of information from a global workspace to specialized processors. In tholonic terms, the global workspace is N: the emergent state that coordinates between D (top-down attention and gating) and C (bottom-up sensory and associative inputs). The workspace becomes unconstrained when

D weakens, consistent with the psychedelic finding of increased global integration (more cross-network connectivity) under DMN suppression [Daw22].

Integrated Information Theory [Ton08] proposes that consciousness is identical to the integrated information (Φ) of a system. High Φ requires both integration (interactions across the system, corresponding to C) and differentiation (a large number of distinct states, which requires D to organize). Under the tholonic model, Φ is maximized when D and C are approximately balanced: too much D produces stereotyped, low-differentiation states; too little D produces high-differentiation but poorly integrated states. The anesthesia data [Cas13] show Φ collapse under both extremes, consistent with the tholonic prediction.

11.2 Limitations

The D and C proxies proposed here (DMN functional connectivity and sensory network / prediction error activity) are population-level measures derived from fMRI. Individual variation in baseline DMN architecture, prior experience with meditation or psychedelics, and set and setting introduce substantial noise into any single-subject B-score estimate.

The B-scores assigned to specific consciousness states in §5 are estimates based on the relative magnitudes reported in the neuroimaging literature, not precisely measured values. They are order-of-magnitude placeholders intended to illustrate the qualitative structure of the state-space. Precise measurement would require a standardized D/C proxy protocol validated against a full range of consciousness states.

The analysis of psychosis in §8 is structural and does not account for the heterogeneity of psychotic disorders, which span schizophrenia, bipolar disorder with psychosis, substance-induced psychosis, and others, each with distinct neurobiological profiles.

12 Conclusions

The Default Mode Network is the brain’s D-apparatus. This is not a metaphor: it is a direct consequence of the DMN’s computational role as the primary maintainer of top-down predictive priors in a hierarchical predictive coding architecture. Psychedelic compounds are D-weakening agents at the network level (through DMN suppression) and at the cellular level (through 5-HT_{2A}-mediated amplification of prediction error propagation). Ego dissolution is N-dissolution: the consequence of D falling below the threshold at which it can maintain the self-model.

The tholonic framework unifies this body of evidence into a single quantitative state-space. Different states of consciousness correspond to different positions in D/C space, with B-score contours marking qualitative transitions. The therapeutic window for psychedelic-assisted therapy, maladaptive high-D states, the phenomenological distinction between psychedelic and psychotic states, and the mechanism of meditation-based D-modulation all follow as structural consequences of the same framework.

The four falsifiable predictions in §10 can be tested against existing neuroimaging datasets from psychedelic clinical trials, resting-state fMRI cohorts of meditators, and depression treatment registries. No new experiments are required for initial validation.

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